WORKED SOLUTION

STEP I 1987, Question 8

Pure • Integration by Substitution

STEP I 1987, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q8

The key idea

A substitution is only a licence to change variables if the new variable moves smoothly and in one direction through an interval as the old one does. Here $t = 1/x$ blows up at $x = 0$, which sits right in the middle of $[-1, 1]$, so the "new limits" -1 and 1 describe a completely different set of t-values. A quick look at the signs of the two integrands confirms something has gone wrong.

Solution

Why the substitution fails

Formally, with $x = 1/t$ we have $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$ and

$$\frac{1}{(1+x^2)^2} = \frac{1}{\left(1+\frac{1}{t^2}\right)^2} = \frac{t^4}{(1+t^2)^2}, \qquad \text{so} \qquad \frac{\mathrm{d}x}{(1+x^2)^2} = \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t,$$

and $x = \pm 1$ corresponds to $t = \pm 1$. This is how one would "show" that the integrals are equal. But it can't be right: the first integrand is positive everywhere, so the first integral is positive, while the second integrand is ≤ 0 everywhere, so the second integral is negative.

? **Ask yourself** As x runs from -1 to 1, which values does $t = 1/x$ actually take?

As x goes from -1 up to 0^-, t goes from -1 down to $-\infty$; at $x = 0$ there is no t at all; and as x goes from 0^+ to 1, t comes down from $+\infty$ to 1. So the x-interval $[-1, 1]$ corresponds to $t \in (-\infty, -1] \cup [1, \infty)$, the *outside* of $[-1, 1]$, not $[-1, 1]$ itself. The substitution rule needs $x = \phi(t)$ with ϕ differentiable on the whole t-interval between the new limits, and $\phi(t) = 1/t$ is not even defined at $t = 0 \in [-1, 1]$. The values of t in $(-1, 1)$ correspond to $|x| > 1$, which lie outside the original range.

Used correctly, on each half separately, the substitution gives for example

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^2)^2} = \int_\infty^1 \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t = \int_1^\infty \frac{t^2}{(1+t^2)^2}\,\mathrm{d}t,$$

which is a true statement (we check it below), but it says nothing about the integral over $[-1,1]$ in t.

The first integral

Put $x = \tan\theta$, so $\mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta$ and $1+x^2 = \sec^2\theta$; this substitution is fine, since tan is smooth and increasing on $[-\frac{\pi}{4}, \frac{\pi}{4}]$. Then

$$\begin{aligned}\int \frac{\mathrm{d}x}{(1+x^2)^2} &= \int \frac{\sec^2\theta}{\sec^4\theta}\,\mathrm{d}\theta = \int \cos^2\theta\,\mathrm{d}\theta = \int \tfrac{1}{2}(1+\cos 2\theta)\,\mathrm{d}\theta \\ &= \tfrac{1}{2}\theta + \tfrac{1}{4}\sin 2\theta = \tfrac{1}{2}\theta + \tfrac{1}{2}\cdot\frac{\tan\theta}{1+\tan^2\theta} = \tfrac{1}{2}\arctan x + \frac{x}{2(1+x^2)},\end{aligned}$$

using $\sin 2\theta = 2\sin\theta\cos\theta = 2\tan\theta\cos^2\theta$. Hence

$$\int_{-1}^1 \frac{\mathrm{d}x}{(1+x^2)^2} = \left[\tfrac{1}{2}\arctan x + \frac{x}{2(1+x^2)}\right]_{-1}^1 = 2\left(\frac{\pi}{8}+\frac{1}{4}\right) = \boxed{\frac{\pi}{4}+\frac{1}{2}}.$$

The second integral

Write the numerator as $-t^2 = 1-(1+t^2)$:

$$\frac{-t^2}{(1+t^2)^2} = \frac{1}{(1+t^2)^2} - \frac{1}{1+t^2}.$$

The first piece integrates to $\frac{\pi}{4}+\frac{1}{2}$ over $[-1,1]$, by the previous part, and $\int_{-1}^1 \frac{\mathrm{d}t}{1+t^2} = [\arctan t]_{-1}^1 = \frac{\pi}{2}$. So

$$\int_{-1}^1 \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t = \frac{\pi}{4}+\frac{1}{2}-\frac{\pi}{2} = \boxed{\frac{1}{2}-\frac{\pi}{4}},$$

which is negative (≈ -0.285), as the sign of the integrand demands, while the first integral is about 1.285.

Checks

- Simpson's rule gives 1.2853981… and −0.2853981… for the two integrals, agreeing with $\frac{\pi}{4}+\frac{1}{2}$ and $\frac{1}{2}-\frac{\pi}{4}$.
- The two answers add up to 1. There's a reason: $\frac{1}{(1+x^2)^2} - \frac{x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$, which is the derivative of $\frac{x}{1+x^2}$, and $\left[\frac{x}{1+x^2}\right]_{-1}^1 = 1$.
- The correct half-range statement checks out: $\int_1^\infty \frac{t^2}{(1+t^2)^2}\,\mathrm{d}t = \frac{\pi}{4} - \left(\frac{\pi}{4}-\frac{\pi}{8}-\frac{1}{4}\right) = \frac{\pi}{8}+\frac{1}{4} = \int_0^1 \frac{\mathrm{d}x}{(1+x^2)^2}$.

Common slips

- Saying only "$1/t$ is undefined at $t = 0$" without explaining that the x-range $[-1, 1]$ corresponds to $|t| \geq 1$: the real point is that the two integrals cover different sets of values.
- Getting the sign of $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$ wrong, which hides the tell-tale sign contradiction.
- Forgetting the $\frac{1}{4}\sin 2\theta$ term when integrating $\cos^2\theta$.

Beyond STEP

The substitution rule $\int_a^b f(\phi(t))\phi'(t)\,\mathrm{d}t = \int_{\phi(a)}^{\phi(b)} f(x)\,\mathrm{d}x$ is proved in a first analysis course from the chain rule and the Fundamental Theorem of Calculus, and the proof shows exactly what is needed: ϕ must be continuously differentiable on all of $[a, b]$. Here it fails at one point, and that one point is enough to wreck the result. Abbott's *Understanding Analysis* develops the Riemann integral and these theorems with care.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.